

MIET2572 – Wearable Systems Workshop #3

Week 7: From Idea to Measurement

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Objective

In this lab you will move from a project idea to a measurable signal.

Key idea: If the signal does not capture the event, the project does not work.

Work must be coordinated through structured meetings.

Expectation

This is an exploratory session.

Your goal is not to get a perfect result.

Your goal is to test, observe, and decide.

If the signal does not work, that is valuable.

System Reminder

Body → Sensor → Signal → Model → Index

You are not building a device. You are defining a measurement.

Task 1 – Define the Measurement

Action

- Define your project in one sentence:

“We measure ____ as ____”

Question

- What physical event are you trying to capture?

Task 2 – Smartphone Placement

Action

- Test 1–2 smartphone placements
- Perform a short movement (2–3 trials)

Trade-offs

Engineering involves trade-offs.

The best signal is not always the best design.

For example, a step counter uses the pocket — not the point of maximum acceleration.

Do not try to optimise everything.

Decide. Move forward. Then re-evaluate.

Analyse

- Is the signal stable?
- Does the placement capture the movement clearly?

Decision

- Select ONE placement
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Task 3 – Pilot Data Collection

Action

- Record data from 1 participant
- Perform 2–3 trials
- Export CSV

Question

- Is your data usable?
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Task 4 – Visualise the Signal

Action

- Load CSV into the notebook
- Plot acceleration vs time

Analyse

- Can you see the event in the signal?
- Is the signal noisy or clear?

Question

- Where is the feature of interest?
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Task 5 – Define the Index

Action

- Define how you will extract your index

Index = feature extracted from signal

Examples

- Peak acceleration
- Time to peak
- Threshold crossing

Question

- What assumption defines your index?

This is a first definition. It will be refined in Workshop 4.

Task 6 – Validate the Signal

Action

- Repeat the movement
- Compare signals across trials

Analyse

- Is the signal repeatable?
- Does the feature appear consistently?

If the signal is not repeatable, the index is not reliable.

This is a first validation. It will be challenged in Workshop 4.

System-Level Reflection

Action

- Identify the weakest part of your system

Analyse

- Is the limitation due to placement, signal, or model?
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Results Log

Trial	Peak / Feature	Comment
1		
2		
3		

Final Reflection

- Does your signal capture the event clearly?
 - Is your index well defined?
 - What is the main limitation?
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Key Takeaway

A project is not an idea. It is a measurable signal.

If the signal does not work, the project does not work.